

Ruoxiang Zhao

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EDUCATION

University of Michigan — Ann Arbor, MI · M.S. in Robotics

Aug 2026 – May 2028 (Expected)

Rensselaer Polytechnic Institute (RPI) — Troy, NY

Aug 2022 – May 2026

B.S. in Mechanical Engineering, Magna Cum Laude · Dean's Honor List (8 semesters)

Coursework: *Mechatronics, Modeling & Control of Dynamic Systems, Vehicle Dynamics, Finite Element Methods, CFD*

SKILLS

Software: Python (NumPy, pandas), C++ (embedded, PlatformIO), MATLAB/Simulink, TypeScript/React/Next.js, LLM & agent APIs (Anthropic, OpenAI), REST APIs, Git, Linux

Robotics & Controls: ROS 2, MPC (CasADi), classical control (PID, feedforward), vehicle dynamics, mechatronics, sensor integration & DAQ, Arduino, ESP32

CAD & Analysis: Siemens NX, SOLIDWORKS, FEA, thermal simulation, tolerance analysis, engineering drawings (GD&T), DFM/DFA

Fabrication: FDM 3D printing (built 2 Voron printers), CNC machining, mill, lathe, welding, rapid prototyping

EXPERIENCE

Stella-Robot — Mechanical Design Intern, Dexterous Hand R&D (Remote)

Jul 2026 – Aug 2026

- Sole structural designer of a hollow-bore, tendon-routed 3-DoF robotic wrist for a humanoid dexterous hand — 2-DoF wrist pitch/yaw plus a wrist-linked thumb-adduction axis enabling high thumb force in a compact package
- Designing the central tendon pass-through (routing paths, guides, anchors) to keep cables pinch-free with stable tension, verifying clearance and interference across the full range of motion
- Quantifying tendon length change vs. wrist pose via multi-pose motion simulation and mechanics calculations; delivering complete 3D CAD, 2D production drawings, and a simulation report on a 5-week cycle

XAL Research Lab, RPI — Undergraduate Researcher, Autonomous Vehicles (Prof. Franck Djeumou)

Feb 2025 – May 2026

- Developed brake-automation, steering, and encoder-based wheel-speed sensing subsystems for a self-driving Can-Am X3 focused on stable control under slipping conditions (ROS 2, Python); presented work to Toyota Research Institute
- Modeled vehicle dynamics and tuned control strategies in MATLAB/Simulink; designed and 3D-printed custom actuator and sensor mounts for the autonomy retrofit

Tesla — Engineering Department Intern

Jul 2025 – Aug 2025

- Reverse-engineered the Model 3 battery-pack coolant distribution manifold — analyzed manufacturing processes and failure modes, redesigned it in 3D CAD for improved thermal dissipation, and validated via simulation
- Designed an integrated mounting bracket and cover for autonomous-driving sensors, prioritizing manufacturability and assembly efficiency (DFM/DFA)

PROJECTS

Automated To-Go Container Dispenser — Senior Capstone, for Sodexo at RPI

Aug 2025 – May 2026

- Led mechanical design of an automated vertical-magazine dispenser (escapement indexer, lead-screw lift, NEMA 23 stepper drive) from concept through fabrication and control
- Built the full software stack solo: RFID student-ID authentication, real-time inventory tracking, usage analytics, automated email alerts, and a Python/Tkinter staff dashboard

TokenJar — ESP32-S3 API Usage Tracker, Hardware + Firmware · github.com/Zhaor3/tokenjar

2026

- Built a desk device showing real-time Anthropic/OpenAI API spend on an ST7789 LCD: C++ firmware (PlatformIO), encoder-driven UI, 24-hour sparkline, REST Admin-API integration, custom 3D-printed enclosure

4-DOF Robotic Arm — Design, Build & Mobile-Platform Integration

Jan 2024 – Aug 2025

- Designed in CAD, sourced components, 3D-printed, and assembled a 4-DOF robotic arm; integrated it onto a tracked mobile-robot platform and engineered it for AI camera-based object detection and manipulation

Cycloidal Drive Gearbox (NEMA 17) — Compact High-Reduction Robot Actuator

Aug 2024 – Jan 2025

- Designed a cycloidal gearbox for compact, high-reduction robotic actuation: full Siemens NX CAD, tolerance analysis, and 3D-printed prototype validation

LEADERSHIP & ACTIVITIES

Rensselaer Student Auto Shop — **Shop Manager** (Sep 2022 – May 2026): ran weekly operations; hands-on support for student vehicle repair

Rensselaer Motorsport (Formula SAE) (Sep 2022 – Dec 2025): wheel-hub CAD; battery-cooling actuator development